PUSH BUTTON START SYSTEM [ADVANCED KEYLESS ENTRY SYSTEM]

id091400111633

Outline

- A push button start system has been adopted that automatically performs authorization of the remote transmitter brought into the vehicle.
- If the engine stalls due to incorrect operation of the clutch by the driver, an engine restarting by depressing clutch pedal function has been adopted to allow the engine to be restarted by only depressing the clutch pedal.(MTX)

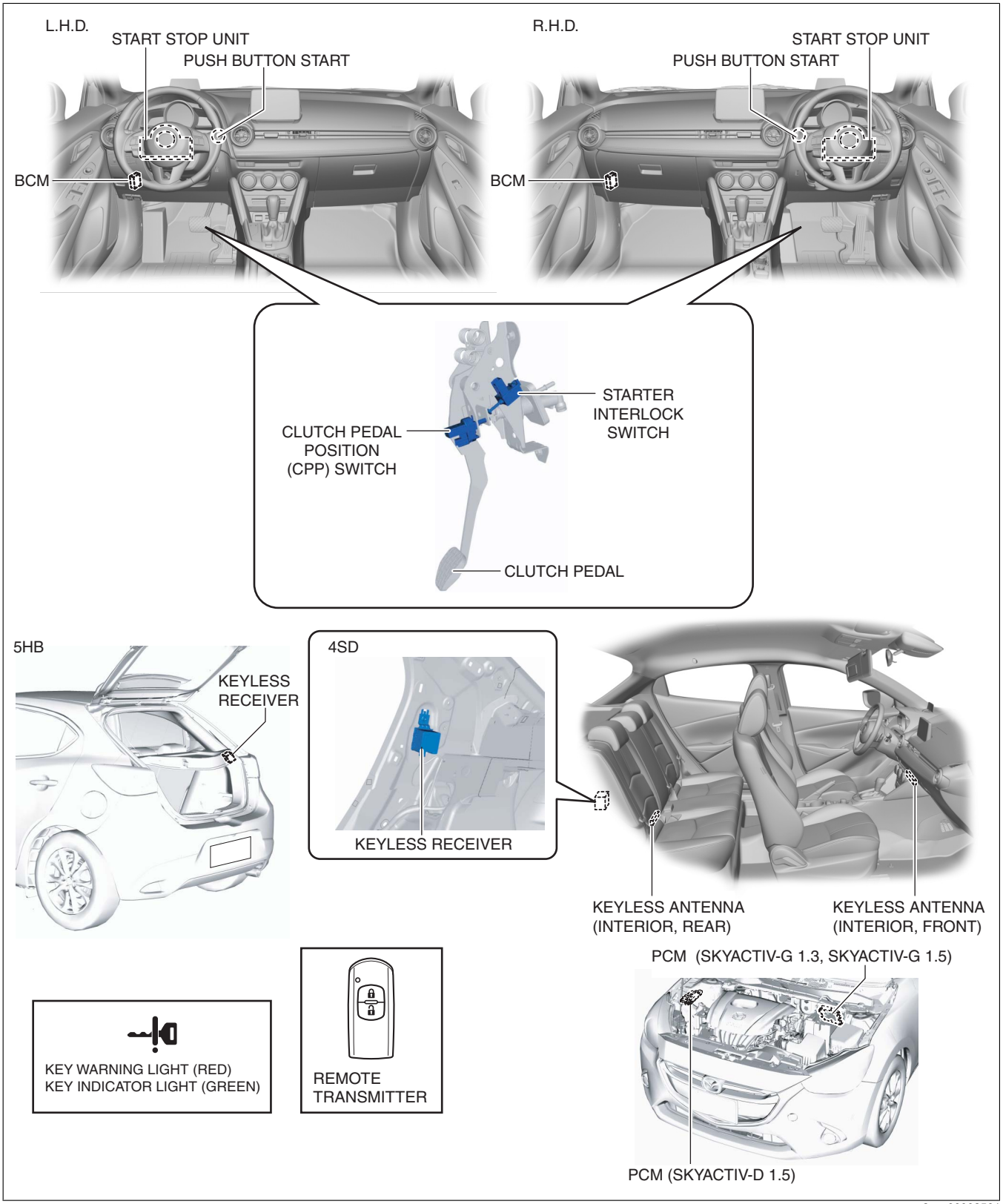
Function

- When the bush button start is pressed, the push button start system performs authorization of the remote transmitter in the vehicle.
  - Engine start: When authorization of the remote transmitter is completed with the engine-start conditions met, the engine starts.
- The guidance function is indicated by the KEY warning light (red) and the keyless warning alarm if there are insufficient conditions for starting the engine, remote transmitter authorization issues, or the steering lock cannot be released.

Engine restarting function by depressing clutch pedal

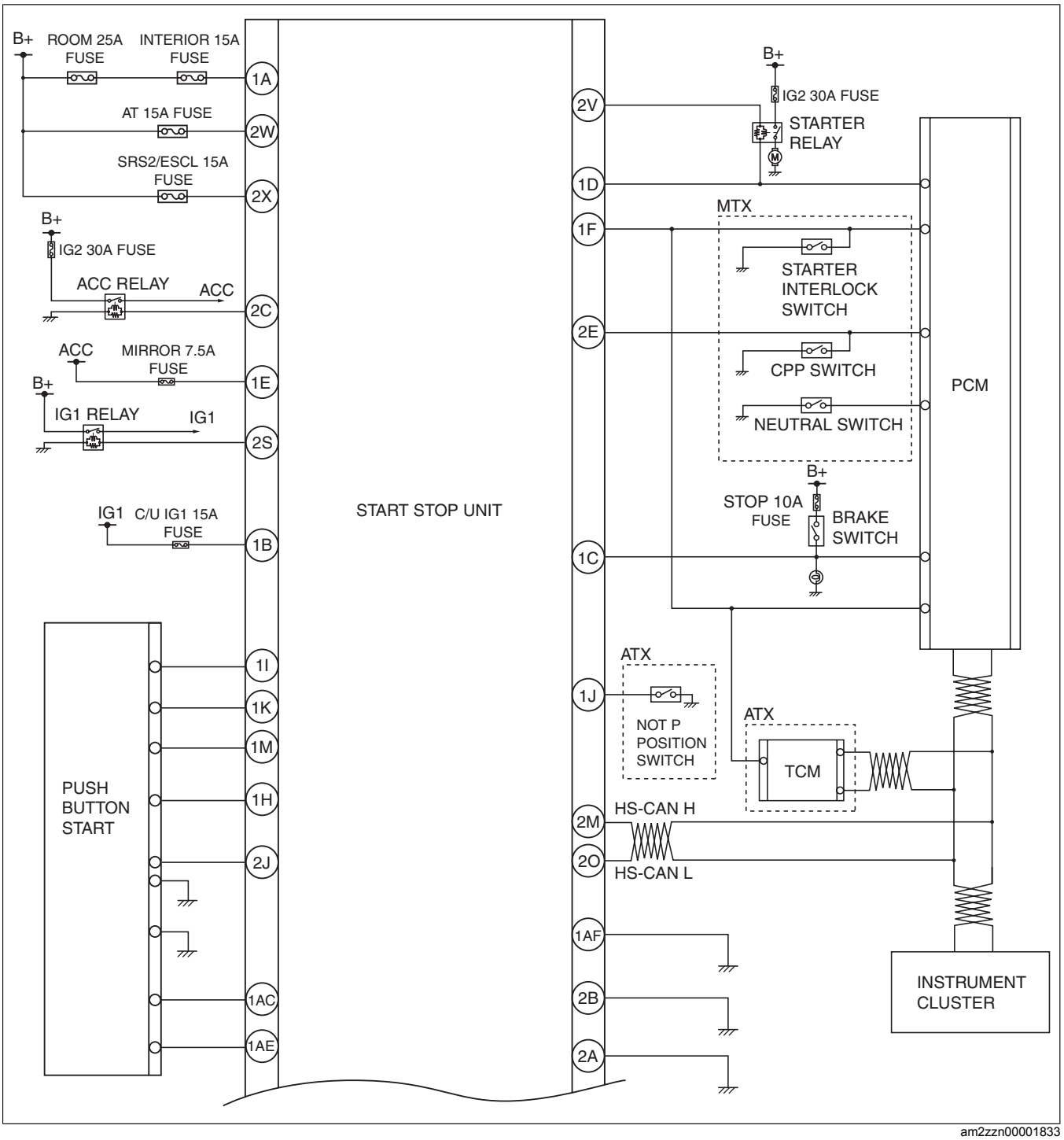
- If the start stop unit determines that the driver intends to restart the engine because the engine has stalled, the engine restarting by depressing clutch pedal function restarts the engine.
- The start stop unit determines that the driver is seated based on the conditions that the driver's seat belt is fastened and the driver's door is closed, and it determines that the driver intends to restart the engine when the clutch pedal is depressed from a released condition **within 3 s** after the engine stalls.

Structural View

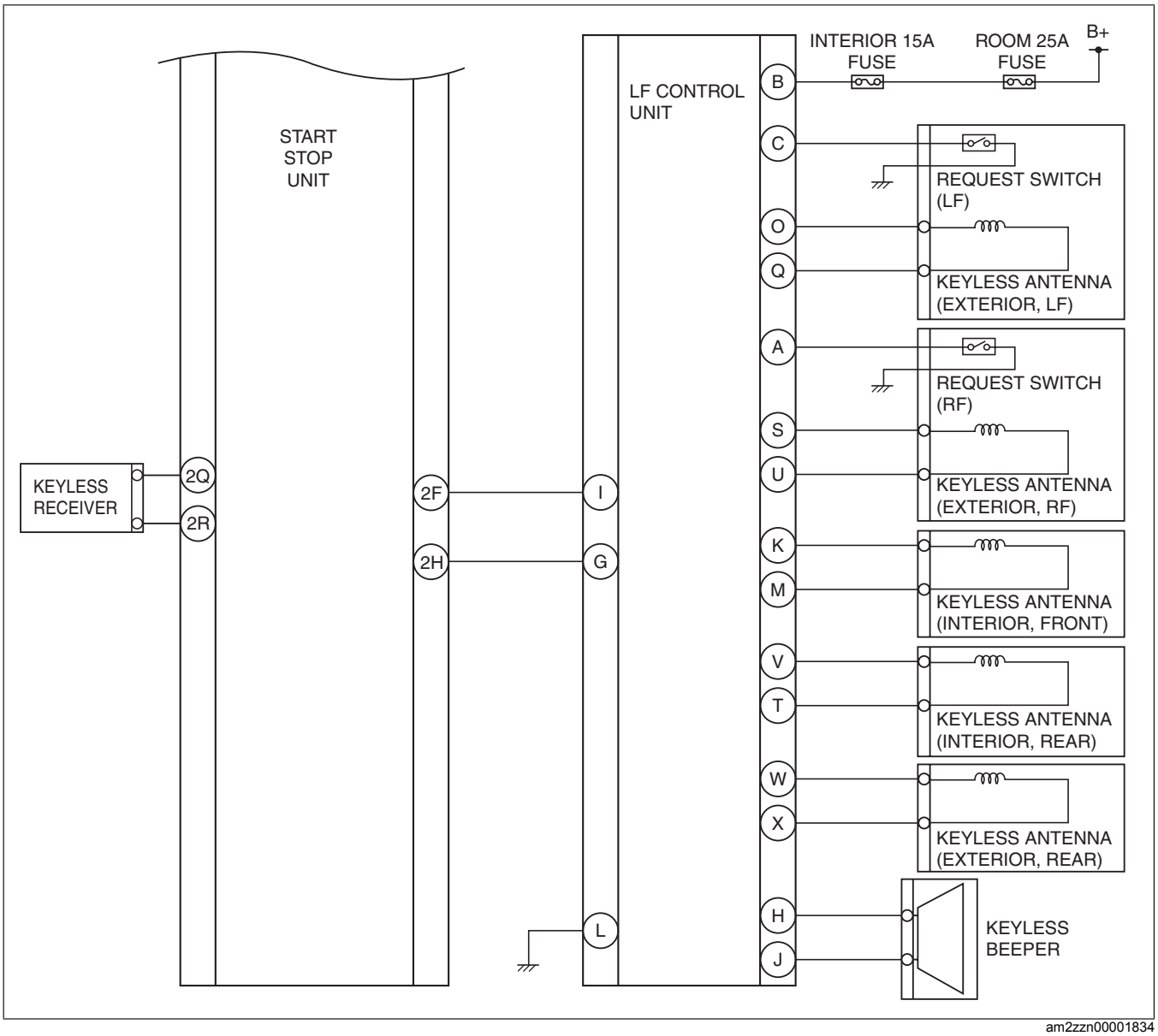


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System Wiring Diagram



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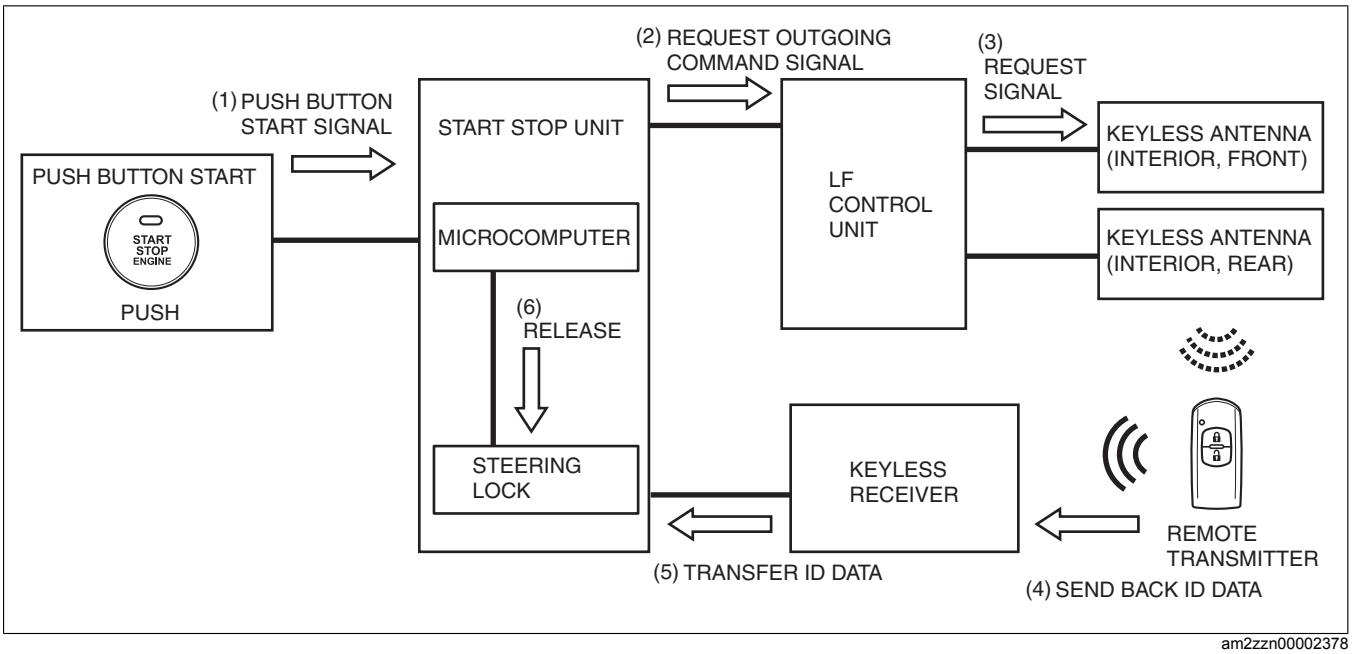


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Operation

Remote transmitter authorization, power supply switching

1. When the push button start is pressed, a push button start signal is input (1) to the start stop unit.
2. When the start stop unit detects a push button start signal, it sends (2) a request outgoing command signal to the LF control unit.
3. When the LF control unit detects the request outgoing command signal, it outputs (3) a request signal using all the keyless antennas in the vehicle interior.
4. The remote transmitter receives the request signal from the interior keyless antennas, and sends (4) ID data to the keyless receiver.
5. The keyless receiver transmits (5) the received ID data to the start stop unit.
6. The start stop unit verifies the ID data and if it matches the steering lock in the module is released (6) and the power supply is switched simultaneously to turn the push button start indicator light (amber) on. For details on the power supply switching, refer to the [POWER SUPPLY SWITCHING]. (See POWER SUPPLY SWITCHING.)



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Engine start

- When all the following conditions are met, authorization of the remote transmitter which is being used to press the push button start is performed to start the engine.
  Engine start conditions
  — Clutch pedal position switch and starter interlock switch on signals (clutch pedal depressed) detected (MTX)
  — Brake switch on signal (brake pedal depressed) detected (ATX)
  — P or N position signal from selector lever (selector lever at P or N position) received (ATX)
  — Remote transmitter is within request signal output range of keyless antennas in vehicle interior

Engine emergency stop

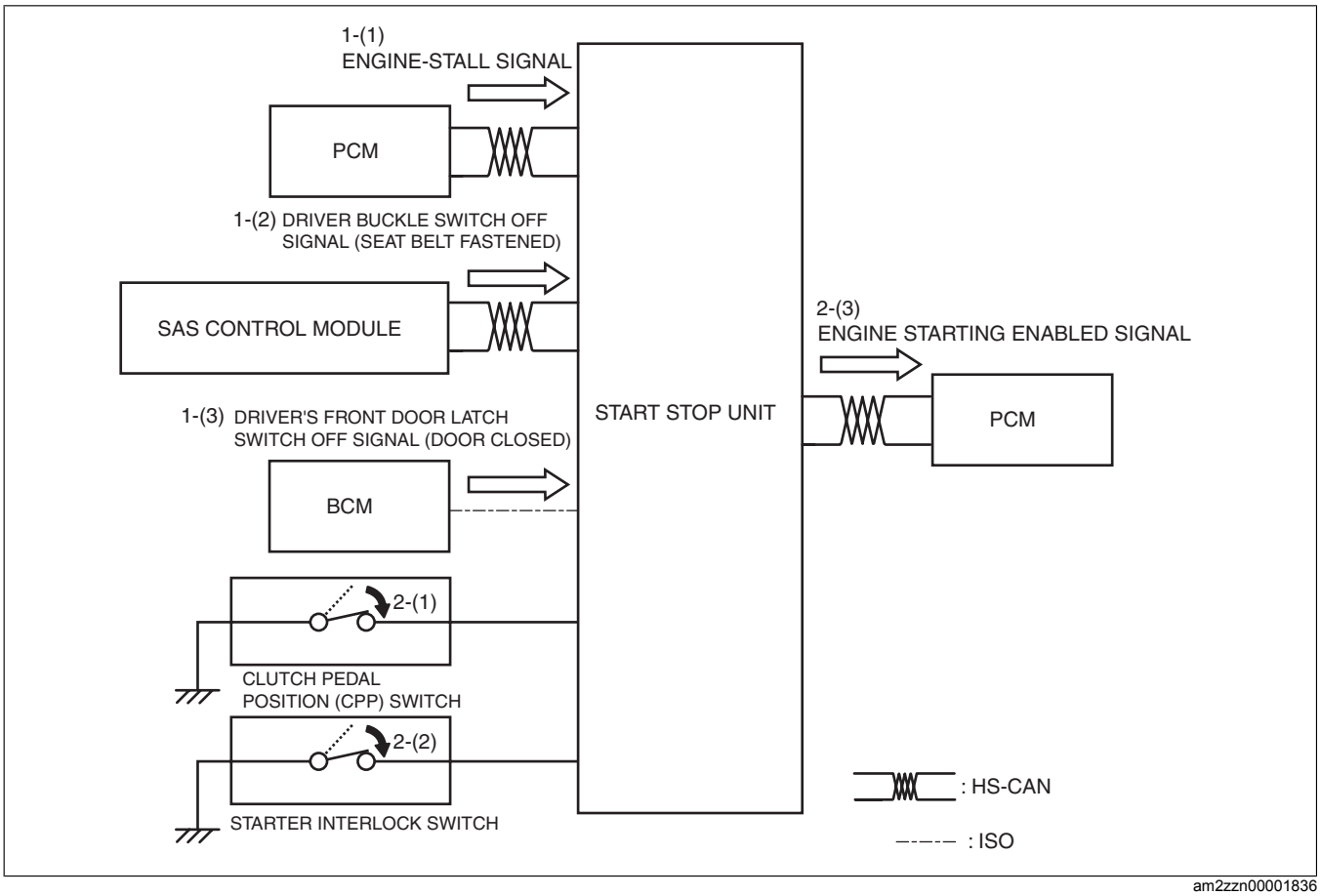
- If a push button start on signal is input **for 3 s or more** (ignition switch held pressed) or a push button start on signal is input 3 times **within 1.5 s** with the engine running, the start stop unit switches the ignition to ACC or OFF according to the position of the shift lever (MTX)/selector lever (ATX).

Back-up mode (MTX) for engine-start

- If the start stop unit determines that either the clutch pedal position switch or starter interlock switch is malfunctioning, the system switches to back-up mode.
- If the push button start is pressed when all the following conditions are met while in the back-up mode, the remote transmitter authorization is performed and the engine starts.
  — Start stop unit detects that either clutch pedal position switch or starter interlock switch is on
  — Neutral switch signal is received via CAN communication from PCM
  — Remote transmitter is within request signal output range of keyless antennas in vehicle interior

Engine restarting function by depressing clutch pedal

1. The start stop unit enables the engine restarting by depressing clutch pedal function operation when all of the following conditions are met.
   — **Within 3 s** after engine-stall signal is received via CAN communication from PCM (1)
   — Driver buckle switch off signal (seat belt fastened) is received via CAN communication from SAS control module (2)
   — Driver door latch off signal (door closed) is received via ISO communication from BCM (3)
2. If all of the following are detected, the start stop unit sends an engine starting enabled signal (3) to the PCM via CAN communication and restarts the engine.
   — Clutch pedal position switch off to on detected (1)
   — Starter interlock switch off to on detected (2)



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Guidance function

Display of methods for resolving issues

- Under the following conditions, the start stop unit sends a warning request signal via CAN transmission to the instrument cluster. It also flashes the push button start indicator (green).
- When the instrument cluster receives the warning request signal under the following conditions, it flashes the KEY warning light (red).

Condition	Instrument cluster		Push button start	
	Keyless warning alarm	KEY warning light (red)	Indicator light (green)	Indicator light (amber)
ID verification is not acceptable	—	Flash	—	Flash
Steering lock cannot be released	Pattern C*1	—	Flash **for 10 s**	—

*1 : For the keyless warning alarm sound pattern, refer to the [KEYLESS WARNING ALARM].(See KEYLESS WARNING ALARM.)